Working Memory in the Chain of Thought Is Not Captured by J-Space

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Abstract

A language model's internal workspace holds at most a few steps of chained reasoning state, at every scale we test, and the chain of thought holds the rest. A common reading of J-space, the concept workspace a fitted Jacobian lens reads from the residual stream, is that reasoning is carried internally and writing relieves the workspace by externalizing state it would otherwise hold. Three measurements contradict the relief picture. Instructed to answer chained arithmetic without writing, models from 1.5B to 671B parameters collapse beyond one to two dependent steps; a 671B model, so instructed, fails two chained additions. Models write every intermediate value from the first step at every difficulty, so externalization begins at no capacity threshold. The allocation between the two sides never moves: damaging the workspace does not increase writing, capping writing does not move computation inward, and ablating the model's active J-space concepts at the strongest dose that preserves ordinary generation changes nothing, even at the one-to-two-step depths where the workspace does perform the arithmetic: the memory the computation uses is not captured by that readout. We then show how this memory works, with causal interventions on six unmodified models (used as released, with no fine-tuning or task-specific training) from four families on arithmetic and narrative state-tracking tasks. Each written value is held at its own token position as a register: overwriting the hidden state there with the state a different number would have produced, text unchanged, redirects the final answer to that never-written number in 0.70 to 1.00 of cases, while size-matched random noise redirects at most 0.01. The register is read through attention to its position (blocking it collapses dependence to at most 0.007 with matched controls unchanged), the read completes by two thirds of the network's depth, and two positions act as independent stores whose planted values compose into a sum appearing nowhere in the input (0.88 to 0.90). For oversight, identical visible text can carry different stored values with different downstream behavior: a chain of thought exposes the symbols of a computation, without guaranteeing their hidden content.

1 Introduction

The chain of thought is the primary working memory of serial reasoning in language models; the internal workspace behind it holds at most a few dependent steps, and writing is not overflow from it. We establish this claim with causal interventions on six unmodified models from four families (Qwen, Phi, OLMo, DeepSeek-R1-Distill), on two task families with exact ground truth for every intermediate value.

The capacity half addresses the overflow picture of externalization: reasoning state is carried internally, and writing takes over when the workspace can no longer hold it. This was our pre-registered

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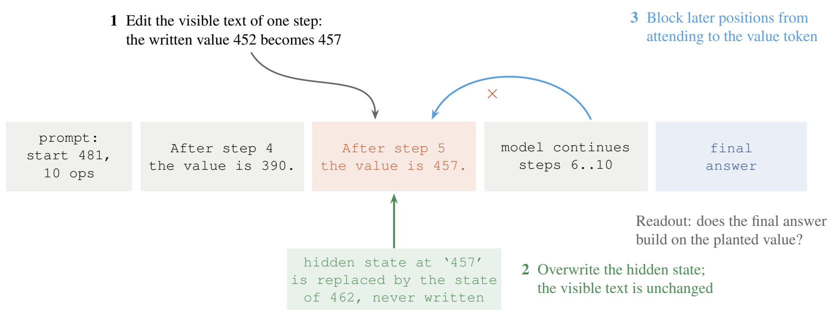

Figure 1: The edit-and-continue protocol. One written step of a correct worked solution is altered and the model continues from it; the readout is whether the final answer builds on the altered value. The interventions: (1) edit the visible text of the step; (2) overwrite the hidden state at the value's token, leaving the text unchanged; (3) block attention from later positions to the value's token.

hypothesis, and it is a natural reading of workspace accounts of reasoning, including J-space, the concept workspace a fitted Jacobian lens reads from the residual stream [Gurnee et al., 2026]. A workspace that writing relieves should show up in three places, and it shows up in none of them.

1. **No fallback.** Instructed to answer chained arithmetic with no writing (token counts confirm the channel is closed), models collapse beyond one to two dependent steps; DeepSeek V3.2, at 671B parameters, answers 0.23 at one step and near zero above (Section 3).
2. **No threshold.** Models write 0.93 to 1.00 of ground-truth intermediates from one-step problems onward, at every scale from 1.5B to 671B (Section 3).
3. **No reallocation.** Damaging the workspace does not increase writing, capping writing does not move computation inward, and ablating the top active J-space concepts at a generation-preserving dose changes neither accuracy nor what gets written (Section 3).

The second half is the mechanism of the memory that remains.

4. **Registers.** The hidden state at a written value's position is a settable value register: planting a never-written value there, text unchanged, redirects the answer at 0.70 to 1.00 across all six models; restoring the correct state under corrupted text reverts it at 0.88 to 1.00; size-matched random noise redirects at most 0.01 on instruction-tuned models (Section 4).
5. **Retrieval.** The register is read through attention to its position: blocking that attention collapses dependence to at most 0.007 on four models in three families, with neighbor, random, and earlier-value blocks at the unblocked level and fluency unchanged. Layer sweeps time the read to the middle third of the network (Section 5).
6. **Composition, and when the register is trusted.** Two positions are independently addressable, composing planted values into their sum (0.88 to 0.90). Models read the register by default; which prompt evidence triggers recomputation is model-specific, and reliance on the trace tracks whether the model can regenerate the state, which reconciles conflicting reports on how trace reliance scales (Sections 4 and 6).

Prior work anticipated the memory role of written tokens theoretically [Merrill and Sabharwal, 2024], behaviorally [Lanchantin et al., 2023], and causally on a fine-tuned model [Shih et al., 2026] and on multiplication and dynamic-programming tasks [Zhu et al., 2025], without testing whether an internal alternative exists, whether the behavior is native to unmodified pretrained models, how the value is retrieved, or whether stores compose.

2 Tasks and protocol

Both task families give exact ground truth for every intermediate value, so the downstream effect of any edit is computable. Arithmetic chains start from a three-digit number and apply d signed two-digit additions and subtractions. Narrative state-tracking problems describe a character whose count

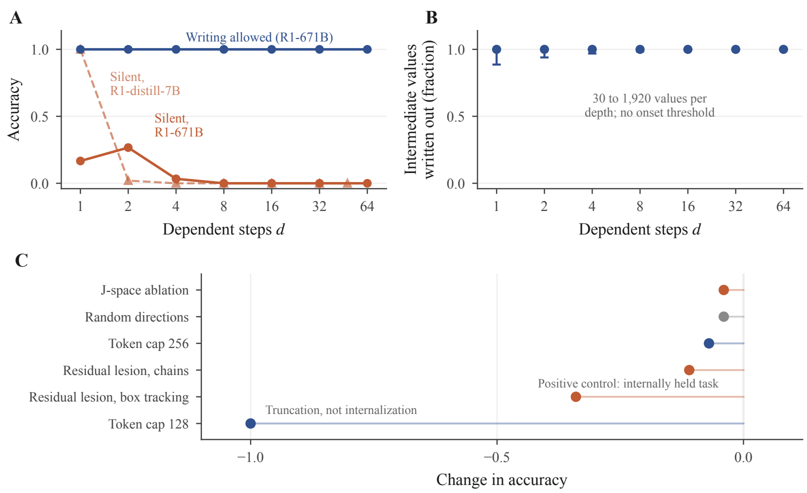

Figure 2: Writing is not triggered by exhausted internal capacity. (A) Instructed to answer without writing, models collapse after one to two dependent steps; with writing allowed, accuracy stays near ceiling (n=30 items per depth). (B) The fraction of intermediate values written is 1.00 at every depth (30 to 1,920 values per depth; 95 percent Wilson intervals), so writing begins at no threshold. (C) No intervention moves state between the workspace and the trace: J-space ablation and random directions leave accuracy unchanged (GSM8K, Qwen3-4B); residual lesions hurt the internally held box-tracking task three times more than the written chain task; a token cap below the per-step floor collapses accuracy by truncation (DeepSeek V3.2, d=16).

of an item changes through varied events, with varied names, items, verbs, and sentence forms. In both, the model writes one line per step giving the running value.

The protocol takes a correct worked solution, changes the value at one step j, truncates immediately after the edited line, and lets the model continue (Figure 1). A no-edit truncation measures the sampling noise floor, 0.00 throughout. Generation uses temperature 0.6 and top-p 0.95 with five samples per item; unparseable outputs are counted separately. Intervals are 95 percent bootstrap intervals over items. White-box experiments use Qwen3-4B, Qwen2.5-7B-Instruct, Phi-3-medium, OLMo-2-7B-Instruct, and DeepSeek-R1-Distill-Qwen 7B and 14B; frontier models (DeepSeek V3.2, Claude Sonnet 4.5, Llama 3.3 70B) are tested behaviorally through assistant prefill. Model revisions, prompts, layer bands, and configurations are in the appendix; code and data will be released.

3 The internal workspace holds a few steps at most

Three measurements bound the internal side of serial reasoning (Figure 2). First, capacity: instructed to output only the final answer, with token counts confirming nothing else was generated, models fail beyond one to two dependent steps (Qwen2.5-7B: 1.00 at $d = 1$, 0.04 at $d = 4$; DeepSeek V3.2 at 671B: 0.23 at $d = 1$, near zero above), and the values used are random, so nothing can be recalled rather than computed. Second, the write policy: models write 0.93 to 1.00 of ground-truth intermediates in free generation from $d = 1$ through $d = 64$, at every scale from 1.5B to 671B, so writing does not begin at any capacity threshold. Third, reallocation: a residual-stream lesion strong enough to cross the accuracy cliff makes traces disintegrate rather than expand; hard token budgets compress wording up to 2.5 times while keeping 0.98 to 1.00 of values, then truncate below roughly twelve tokens per step; and ablating J-space [Gurnee et al., 2026] changes nothing at the strongest dose preserving ordinary generation (calibrated on neutral text: next-token agreement 0.87, perplexity ratio 1.12). The J-space ablation removes the top 16 active concepts at every position and leaves chain-of-thought accuracy, bare-answer accuracy, and the amount written unchanged, on synthetic chains and on GSM8K, doing no more than a size-matched random-directions ablation. The null holds at one to two steps, where models do perform the arithmetic internally.

This bounds the particular readout tested rather than the workspace as a whole: the dose is capped at generation-preserving strength, the lens is one fitted artifact, and a written token's contextual

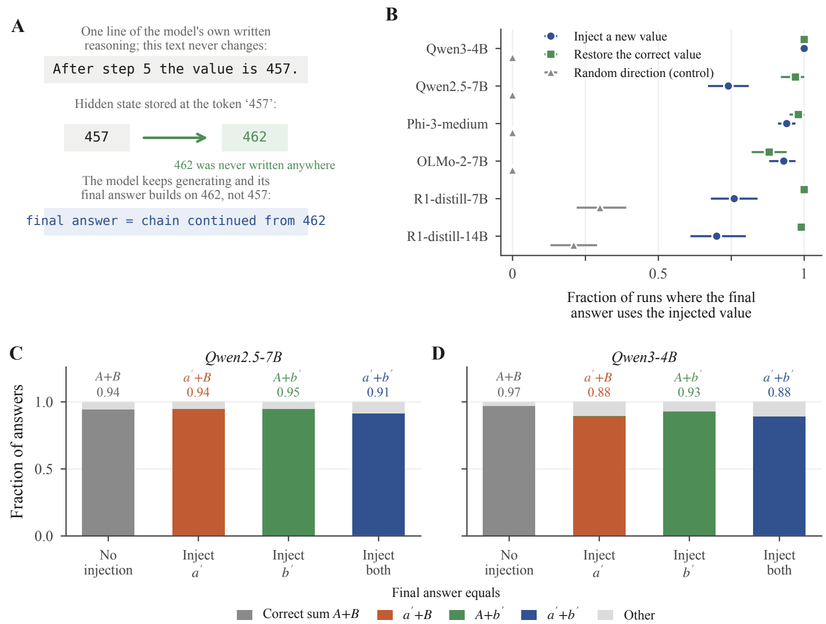

Figure 3: The hidden state under a written value is a settable register, and two registers compose. (A) The text still says 457; the hidden state at that token now encodes 462, which appears nowhere, and the final answer continues from 462. (B) Fraction of runs whose final answer uses the injected value; bars are 95 percent bootstrap intervals over items. The reasoning-distilled models re-solve after their reasoning phase, which raises their random-control rate; re-derivation can only produce the correct answer. (C, D) With two counters A and B, injecting a replacement for either or both (a', b') makes the answer equal the matching sum; answers using only one of two injected values occur at 0.01.

representation and its J-space component are not exclusive alternatives, since the lens samples the same residual states our patches overwrite. The workspace also exceeds J-space by construction: attention and MLP interiors and the KV cache fall outside the lens's domain, and within the residual stream the lens spans a fitted concept dictionary rather than the full state. Our register result is an existence proof of the gap: the planted value is causally decisive, sits in residual states the lens samples, and survives removal of the top 16 concepts.

4 The written position holds a settable value register

Figure 3 decomposes the effect of editing a written value. Restoring the correct hidden state while the text still shows a corrupted value returns the answer to correct in 0.88 to 1.00 of cases. Planting a third value that appears nowhere in the text redirects the answer to it in 0.70 to 1.00 of cases. The size-matched random control sits at or below 0.01 for the four instruction-tuned models. Because different planted values produce answers following those values, the value content is the causal quantity, and because the planted value is a mid-chain intermediate the model never produced, the patch does not inject the answer. The effect persists to 40-step chains (0.98 to 0.95 on Qwen3-4B). Rates are conditioned on items where the text edit was followed (0.44 to 0.98 by model; full table in the appendix). No fine-tuning or prompt engineering is involved; the released models already work this way.

The result is a property of written state, not of one template. On the narrative task the pattern replicates: a planted never-written count is followed at 0.83 [0.75, 0.90] on Qwen3-4B and 0.86 [0.79, 0.92] on Phi-3-medium, restoring the correct state returns the answer at 0.89 and 0.94, and the random control is 0.00 in both ($n = 86$ and 89).

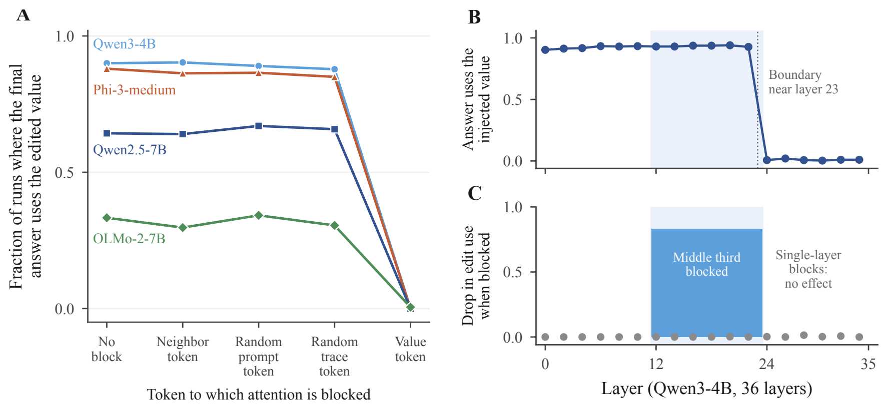

Figure 4: Written values are retrieved through attention to their token, across the middle third of layers. (A) Blocking later positions' attention to the edited value's token drops edit use to zero on four models; matched control blocks change nothing (n=600 per condition per model). (B) Overwriting the hidden state at any single layer up to 22 redirects the answer; from layer 24 on it does nothing (Qwen3-4B, 36 layers, n=60). (C) Blocking attention across layers 12 to 23 (shaded in both panels) removes 0.83 of edit use; no single layer's block matters (gray dots, n=80).

Written positions are independently addressable. In a task with two counters summed at the end (Figure 3C, D), planting a never-written value at one final position yields the corresponding mixed sum (0.94 and 0.93 on Qwen2.5-7B, 0.89 and 0.93 on Qwen3-4B). Planting both yields their joint sum at 0.90 and 0.88, at or above the product of the single-position rates (0.89 and 0.83). The random patch leaves the clean answer in place (0.96 and 0.95), and answers using only one of the two planted values occur at 0.01. This is the result that separates structured memory from a strong perturbation dragging the output: the reported number must be assembled at answer time from two values that were set independently, at different positions, and never written anywhere.

5 Retrieval runs through attention to the written position

Blocking attention from all post-edit positions to the edited value's position collapses the answer's dependence on that value on every model tested (Figure 4A). Matched blocks of the adjacent words, of a random prompt position, or of a random position elsewhere in the trace leave the dependence at its unblocked level. Blocking the operand the next step needs breaks the arithmetic itself, yielding answers that match neither the edit nor the correct value. The collapse holds across three families and on both task formats (natural-language task: 0.730 to 0.007 on Qwen3-4B, 0.720 to 0.003 on Phi-3-medium).

The block removes the value, not the ability to generate. Under the value-position block the unparseable-output rate stays at 0.000 (0.040 on OLMo-2-7B, equal to its unblocked rate). The model produces a fluent, parseable number that matches neither the edited value nor the correct one: it computes forward from a value it can no longer retrieve. Reading is the preferred path rather than the only one: replacing the value's characters with unreadable dots, leaving attention untouched, also removes following, and the Qwen models then reconstruct the value from the previous line in 0.60 to 0.73 of cases while Phi-3-medium never does (0.00).

Retrieval occurs in the middle third of the network. Layer-resolved versions of both interventions locate the read in depth on Qwen3-4B (36 layers; Figure 4B, C). Overwriting the value's hidden state at any single layer from 0 through 22 redirects the answer at 0.90 to 0.94, matching the full multi-layer patch (0.92), while single-layer patches at layer 24 and beyond redirect at 0.02 or less (n = 60): the register can be rewritten anywhere upstream of its consumers and nowhere downstream of them. Blocking the value-position attention only at the middle third of layers (12 through 23) collapses following from 0.857 to 0.025, blocking only the early or late third leaves it unchanged (0.857 and 0.863), and no single layer's block matters (0.84 to 0.89 at every layer, n = 80): the read is distributed redundantly across the middle third's layers and confined to it. The two sweeps agree on the same boundary, near layer 23. Coarse sweeps on two further models (appendix) show the structure generalizes while the boundary's depth is model-specific: on Phi-3-medium, early- and mid-third patches redirect at 0.89 and 0.87 and late-third patches at 0.007, matching the Qwen3-4B shape, while on Qwen2.5-7B patches in every third redirect at 0.70 to 0.73, so its reads extend into the final third of its 28 layers.

6 When the written value is read versus recomputed

Models read the register by default. On DeepSeek V3.2, an edited value the prompt merely implies is followed at 0.93, while an edited value the prompt generatively defines (the prompt states the two numbers whose sum becomes the running value) is reverted at 0.43; with the defining sentence present but the edit two steps away, following returns to 0.96, so the reversion targets the defined value. Claude Sonnet 4.5 recomputes more broadly, reverting most when a stated checkpoint is the value's only source. Llama 3.3 70B follows edits at 0.97 whether or not the value is prompt-defined, despite solving the same chains from scratch, so the capacity to recompute does not imply recomputing. Qwen models at 4B and 7B never revert. These conditions come from separate experiments, so we claim within-experiment contrasts only; what governs the choice remains open.

The same test bounds the mechanism on natural benchmarks and bears on a disputed question. On GSM8K worked solutions, an edited intermediate is followed at 0.87 by Qwen3-4B and at 0.10 by V3.2, since a benchmark intermediate is often one operation from numbers still in the prompt and a stronger model recomputes it. Within one model, V3.2 follows edits at 0.93 on chains it cannot regenerate and at 0.10 on GSM8K. Lanham et al. [2023] report reliance on written reasoning declining with scale on standard benchmarks, a result later work disputes; our contrasts indicate the direction depends on the state itself. On regenerable state, stronger models depend less on the trace; on state no model can regenerate, dependence is high at every scale we test. Faithfulness of the trace to the computation is a property of the state's regenerability.

7 Related work

Our interventions build on causal tracing and activation patching [Meng et al., 2022] and causal abstraction [Geiger et al., 2021]; Heimersheim and Nanda [2024] caution that broad patches conflate a component mattering with deleting information at a position, which the single-layer sweep in Section 5 addresses. The memory role of written tokens in prior work is discussed in the introduction. Faithfulness studies [Turpin et al., 2023, Lanham et al., 2023] and filler-token results [Pfau et al., 2024] bound our claims to state the model does not regenerate internally.

8 Limitations

Tasks are numeric and count state tracking, so our claims concern serial task state rather than working memory in general; code, plans, and multi-entity state are untested. The J-space result bounds one fitted lens at generation-preserving doses and does not measure the J-space content of the written positions themselves. The fine layer sweeps are established on one model; the cross-model sweeps are coarse, and on Qwen2.5-7B they do not resolve the read boundary. White-box evidence is at 4B to 14B; frontier evidence is behavioral. The only patch control is size-matched noise; a structured control planting a valid activation for a different quantity would further isolate value content.

9 Conclusion

Serial reasoning state outgrows a language model's internal workspace within one to two dependent steps at every scale tested, writing begins at no threshold, and no intervention shifts state between the workspace and the trace, so the chain of thought is working memory rather than overflow. It works as compositional registers: settable by intervention, retrieved through attention in the middle layers, independently addressable, and decisive for the answer. For oversight the implication cuts both ways: the written positions are a causally used memory channel, and identical text can carry different stored values, so trace monitoring observes the channel without verifying the state.

References

Atticus Geiger, Hanson Lu, Thomas Icard, and Christopher Potts. Causal abstractions of neural networks. In *NeurIPS*, 2021.

Wes Gurnee et al. Reading concepts from the residual stream with jacobian lenses. *arXiv preprint*, 2026.

Stefan Heimersheim and Neel Nanda. How to use and interpret activation patching. *arXiv:2404.15255*, 2024.

Jack Lanchantin, Shubham Toshniwal, Jason Weston, Arthur Szlam, and Sainbayar Sukhbaatar. Learning to reason and memorize with self-notes. In *NeurIPS*, 2023.

Tamera Lanham et al. Measuring faithfulness in chain-of-thought reasoning. *arXiv:2307.13702*, 2023.

Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *NeurIPS*, 2022.

William Merrill and Ashish Sabharwal. The expressive power of transformers with chain of thought. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.07923.

Jacob Pfau, William Merrill, and Samuel R. Bowman. Let's think dot by dot: Hidden computation in transformer language models. In *Conference on Language Modeling (COLM)*, 2024. arXiv:2404.15758.

Andy Shih et al. Do models read what they write? causal registers in scratchpad reasoning. *arXiv:2606.29522*, 2026.

Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. Language models don't always say what they think: Unfaithful explanations in chain-of-thought prompting. In *NeurIPS*, 2023.

Fangwei Zhu et al. Chain-of-thought tokens are computer program variables. *arXiv:2505.04955*, 2025.